

Nomenclature and Equation References

Corrected HVAC reduced-order degradation framework

Editorial note. The table uses unambiguous recommended manuscript notation. In particular, $y_{GHI,t}$ replaces the first use of g_t , nd,t replaces the day-of-year use of d_t , and overbarred objective indicators replace e_t , d_t , c_t , and g_t . This removes symbol collisions in the earlier equation draft. Units marked “–” are dimensionless.

1. Nomenclature

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
Indices and operators	t	Simulation time-step index	–	5–61	
Indices and operators	k	Stochastic-search iteration index	–	52–55	
Indices and operators	i	Candidate-solution index	–	53–55	
Indices and operators	j	Energy component index: HVAC, fan, pump, or auxiliary	–	41–43	
Indices and operators	m	Calendar month index	–	26–29, 38, 41	
Indices and operators	r_t	Rule-based operational-state label assigned at time step t	–	42	
Indices and operators	Δt	Simulation time-step duration	h	22–25, 43, 49	
Indices and operators	$\tau = \Delta t/24$	Simulation time-step duration expressed in days	day	30–31	
Indices and operators	$\text{clip}(x,l,u)$	Clipping operator that limits x to the interval $[l,u]$	–	10, 24, 27, 38, 41–42, 53	
Indices and operators	$\max(\cdot), \min(\cdot)$	Maximum and minimum operators	–	7, 9–11, 15, 23, 25, 29, 32, 34–35, 38–40, 54	
Indices and operators	$\lfloor \cdot \rfloor$	Floor operator	–	54	
Indices and operators	$\odot$	Element-wise vector multiplication	–	53	
Indices and operators	$\mathcal{N}(0,I)$	Multivariate standard normal distribution	–	53	
Indices and operators	I	Identity matrix in the Gaussian candidate distribution	–	53	
Indices and operators	$\mathcal{E}_k$	Elite candidate set at search iteration k	–	55	
Indices and operators	$(\cdot)^+$	State immediately after maintenance restoration	–	58	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
Indices and operators	(·)raw, (·)(s), (·)final	Raw, seasonally corrected, and final operational-state-corrected quantities	–	41–43	
Building and design quantities	A	Conditioned floor area	m ²	1–8, 19, 21, 40	
Building and design quantities	q ^c ,A	Area-normalized design cooling intensity	W m ⁻²	1	
Building and design quantities	q ^h ,A	Area-normalized design heating intensity	W m ⁻²	2	
Building and design quantities	Q ^c ,des	Nominal/design cooling capacity	kW	1, 12–15, 20, 23, 25, 28, 46, 49	
Building and design quantities	Q ^h ,des	Nominal/design heating capacity	kW	2, 16, 20, 23, 25, 28	
Building and design quantities	v _{air} ,A	Area-normalized nominal airflow rate	m ³ h ⁻¹ m ⁻²	3	
Building and design quantities	V _{air,nom}	Nominal HVAC airflow rate	m ³ h ⁻¹	3, 37, 39	
Building and design quantities	N _{max}	Maximum occupant count	person	4–5	
Building and design quantities	ρ _{occ}	Occupant density	person m ⁻²	4	
Building and design quantities	V	Equivalent conditioned air volume	m ³	21	
Building and design quantities	h _f	Floor-to-floor height used to estimate V = Ah _f	m	21	
Building and design quantities	ACH	Air infiltration rate	h ⁻¹	21	
Occupancy and gains	o _t	Calendar- and hour-dependent occupancy fraction	–	5, 7–8, 27, 38, 40, 45	
Occupancy and gains	N _t	Effective number of occupants at time step t	person	5	
Occupancy and gains	PL	Installed lighting electrical power	kW	6–7	
Occupancy and gains	PE	Installed equipment electrical power	kW	6–7	
Occupancy and gains	p _L ,A	Area-normalized lighting power density	W m ⁻²	6	
Occupancy and gains	p _E ,A	Area-normalized equipment power density	W m ⁻²	6	
Occupancy and gains	f _{use}	Occupancy-dependent internal-gain use factor	–	7	
Occupancy and gains	Q _{int,t}	Internal electrical sensible gain	kW	7, 15, 17	
Occupancy and gains	Q _{sens}	Sensible heat released per person	W person ⁻¹	8	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
Occupancy and gains	$Q_{occ,t}$	Sensible occupant heat gain	kW	8, 15, 17	
Weather and static loads	$T_{a,t}$	Outdoor dry-bulb temperature	°C	9, 22, 34–35, 45	
Weather and static loads	$T_{sp,t}$	Controlled zone temperature setpoint	°C	9, 45, 48, 52	
Weather and static loads	$\Delta T_{c,t}$	Positive cooling temperature difference	K	9, 12, 14	
Weather and static loads	$\Delta T_{h,t}$	Positive heating temperature difference	K	9, 16	
Weather and static loads	GHI_t	Global horizontal solar irradiance	W m ⁻²	10	
Weather and static loads	RH_t	Outdoor relative humidity	%	10, 45	
Weather and static loads	γGHI_t	Normalized solar-irradiance modifier; recommended notation replacing the duplicated symbol g_t	–	10, 13	
Weather and static loads	μRH_t	Humidity multiplier applied to the static cooling load	–	10, 15	
Weather and static loads	kRH	Cooling-load humidity multiplier coefficient	% ⁻¹	10	
Weather and static loads	nd_t	Day-of-year index; recommended notation replacing d_t to avoid collision with the normalized degradation objective	day	11	
Weather and static loads	s_t	Daily solar-season modifier	–	11, 13	
Weather and static loads	$\chi_{env}, \chi_{sol}, \chi_{inf}$	Binary activation switches for envelope, solar, and infiltration terms	–	12–16	
Weather and static loads	ac, ah	Reduced-order envelope cooling and heating coefficients	–	12, 16, 20	
Weather and static loads	$\Delta T_{c,ref}$	Reference cooling temperature difference	K	12, 14, 20	
Weather and static loads	$\Delta T_{h,ref}$	Reference heating temperature difference	K	16, 20	
Weather and static loads	f_{sol}	Solar cooling-load coefficient	–	13	
Weather and static loads	$f_{inf,c}, f_{inf,h}$	Cooling and heating infiltration coefficients	–	14, 16	
Weather and static loads	$Q_{c,env,t}$	Envelope component of static cooling load	kW	12, 15	
Weather and static loads	$Q_{c,sol,t}$	Solar component of static cooling load	kW	13, 15, 22	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
Weather and static loads	$Q_{c,inf,t}$	Infiltration component of static cooling load	kW	14–15	
Weather and static loads	$Q_{lat,t}$	Latent cooling load; zero when latent coupling is disabled	kW	15	
Weather and static loads	$Q_{c,stat,t}$	Total static cooling load before dynamic RC blending	kW	15, 25	
Weather and static loads	$Q_{h,env,t}$	Envelope component of static heating load	kW	16, 18	
Weather and static loads	$Q_{h,inf,t}$	Infiltration component of static heating load	kW	16, 18	
Weather and static loads	f_{credit}	Fraction of internal gains credited against heating demand	–	17	
Weather and static loads	$Q_{credit,t}$	Internal-gain heating credit	kW	17–18	
Weather and static loads	$Q_{h,stat,t}$	Static heating demand before dynamic RC blending	kW	18, 25	
Dynamic RC model	C_z	Equivalent zone-air thermal capacitance	kWh K ⁻¹	19, 22–24	
Dynamic RC model	C_m	Equivalent building-mass thermal capacitance	kWh K ⁻¹	19, 24	
Dynamic RC model	c_z, c_m	Area-normalized zone-air and mass heat capacities	kWh m ⁻² K ⁻¹	19	
Dynamic RC model	H_{env}	Effective envelope heat-transfer coefficient	kW K ⁻¹	20, 22, 24	
Dynamic RC model	f_{UA}	Multiplier applied to the effective envelope UA term	–	20	
Dynamic RC model	H_{vent}	Ventilation/infiltration heat-transfer coefficient	kW K ⁻¹	21–22	
Dynamic RC model	ρ_{air}	Air density	kg m ⁻³	21	
Dynamic RC model	$c_{p,air}$	Specific heat capacity of air	kJ kg ⁻¹ K ⁻¹	21	
Dynamic RC model	f_{vent}	Ventilation heat-transfer multiplier	–	21	
Dynamic RC model	H_{mz}	Thermal coupling coefficient between building mass and zone air	kW K ⁻¹	21–24	
Dynamic RC model	h_{mz}	Area-normalized mass-zone coupling coefficient	W m ⁻² K ⁻¹	21	
Dynamic RC model	$T_{z,t}$	Equivalent zone-air state temperature	°C	22, 24	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
Dynamic RC model	$T_{m,t}$	Equivalent building-mass state temperature	°C	22, 24	
Dynamic RC model	$T_{z,t,free}$	Predicted free-floating zone temperature before HVAC action	°C	22–24	
Dynamic RC model	$T_{h,sp}, T_{c,sp}$	Heating and cooling thermostat setpoints used by the dynamic solver	°C	23	
Dynamic RC model	$Q_{sol,a,t}, Q_{sol,m,t}$	Solar gains assigned to zone air and thermal mass	kW	22, 24	
Dynamic RC model	$Q_{int,a,t}, Q_{int,m,t}$	Internal gains assigned to zone air and thermal mass	kW	22, 24	
Dynamic RC model	$Q_{c,dyn,t}, Q_{h,dyn,t}$	Dynamic cooling and heating demands required to enforce thermostat limits	kW	23–25	
Dynamic RC model	ϵ	Dynamic control effectiveness	–	24	
Dynamic RC model	T_{min}, T_{max}	Lower and upper bounds for dynamic thermal states	°C	24	
Dynamic RC model	β_{RC}	Fraction replacing static load with the dynamic RC load	–	25	
Dynamic RC model	$Q_{c,RC,t}, Q_{h,RC,t}$	Cooling and heating loads after static/dynamic blending	kW	25–26	
Monthly and calibration layers	ac,m, ah,m	Month-specific cooling and heating availability factors	–	26, 28	
Monthly and calibration layers	$Q_{c,av,t}, Q_{h,av,t}$	Loads after monthly availability correction	kW	26, 29	
Monthly and calibration layers	w_o	Weight of occupancy in the minimum-load factor	–	27	
Monthly and calibration layers	$f_{occ,t}$	Occupancy factor used for minimum operational loads	–	27–28	
Monthly and calibration layers	rc,m, rh,m	Month-specific minimum cooling and heating load fractions	–	28	
Monthly and calibration layers	$Q_{c,min,t}, Q_{h,min,t}$	Month-specific minimum operational loads	kW	28–29	
Monthly and calibration layers	$Q_{c,op,t}, Q_{h,op,t}$	Accepted operational loads before capacity limiting	kW	29	
Degradation model	Rf_t	Heat-exchanger fouling resistance at time step t	$m^2 K W^{-1}$	30, 33–35, 56, 58	
Degradation model	Rf^*	Asymptotic/reference fouling resistance	$m^2 K W^{-1}$	30, 33–35	
Degradation model	Bf	Fouling growth-rate coefficient	day^{-1}	30	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
Degradation model	md_t	Accumulated filter dust mass	kg	31–32, 58	
Degradation model	$\dot{m}d$	Filter dust accumulation rate at unit airflow fraction	kg day ⁻¹	31	
Degradation model	α_t	Normalized airflow/control fraction	–	31, 37–39, 45, 48, 52	
Degradation model	ΔP_t	Filter/system air-side pressure drop used by the degradation model	Pa	32–33, 37, 39, 57	
Degradation model	ΔP_{clean}	Clean-filter pressure drop	Pa	32	
Degradation model	ΔP_{max}	Maximum pressure-drop cap	Pa	32–33	
Degradation model	k_{clog}	Pressure-drop increase per unit accumulated dust mass	Pa kg ⁻¹	32	
Degradation model	δ_t	Implemented composite degradation indicator	–	33, 40, 45, 49–51, 56–57	
HVAC performance and power	$COP_{c,nom}, COP_{h,nom}$	Nominal cooling and heating coefficients of performance	–	34–35	
HVAC performance and power	$COP_{c,t}, COP_{h,t}$	Degradation- and temperature-dependent cooling and heating COP	–	34–36	
HVAC performance and power	k_{age}	Annual COP-aging coefficient	year ⁻¹	34–35	
HVAC performance and power	y_t	Elapsed simulated age	year	34–35	
HVAC performance and power	$QH_{VAC,t}$	Active thermal HVAC load under the selected mode	kW	36	
HVAC performance and power	COP_t	COP selected for the active HVAC mode	–	36	
HVAC performance and power	$PH_{VAC,t}$	Thermal HVAC electrical power	kW	36, 43	
HVAC performance and power	η_{fan}	Fan total efficiency	–	37, 39	
HVAC performance and power	$P_{fan,raw,t}$	Raw pressure-flow fan electrical power	kW	37, 39	
HVAC performance and power	$r_{fan,t}$	Schedule-based fan runtime fraction	–	38–39	
HVAC performance and power	r_{min}	Minimum fan runtime fraction	–	38	
HVAC performance and power	w_{occ}, w_{HVAC}	Occupancy and HVAC-activity weights in the fan runtime model	–	38	
HVAC performance and power	$I_{HVAC,t}$	Binary indicator of active thermal HVAC demand	–	38	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
HVAC performance and power	$\alpha_{fan,m}$	Month-specific fan availability factor	–	38	
HVAC performance and power	$\alpha_{min,fan}$	Minimum airflow fraction used by the schedule-based fan model	–	38	
HVAC performance and power	$\alpha_{floor,t}$	Airflow fraction after application of the fan airflow floor	–	38–39	
HVAC performance and power	$P_{fan,des,t}$	Fan design power at the current pressure drop	kW	39	
HVAC performance and power	$P_{fan,sch,t}$	Schedule-based fan power floor	kW	39	
HVAC performance and power	$P_{fan,t}$	Accepted fan electrical power	kW	39, 43	
HVAC performance and power	$f_{op,t}$	Pump/auxiliary operating factor	–	40	
HVAC performance and power	$p_{pump,A}, p_{aux,A}$	Area-normalized pump and auxiliary power densities	W m ⁻²	40	
HVAC performance and power	$P_{pump,t}, P_{aux,t}$	Pump and auxiliary electrical powers	kW	40, 43	
Corrections, energy, and carbon	$F_{j,mraw}$	Raw month-specific calibration factor for component j	–	41	
Corrections, energy, and carbon	λ_s	Damping factor for monthly component calibration	–	41	
Corrections, energy, and carbon	F_{min}, F_{max}	Lower and upper monthly-factor bounds	–	41	
Corrections, energy, and carbon	$F_{j,m}$	Damped and bounded monthly component factor	–	41	
Corrections, energy, and carbon	$G_{j,rraw}$	Raw operational-state factor for component j and regime r	–	42	
Corrections, energy, and carbon	λ_o	Damping factor for operational-state correction	–	42	
Corrections, energy, and carbon	G_{min}, G_{max}	Lower and upper operational-state-factor bounds	–	42	
Corrections, energy, and carbon	$G_{j,r,t}$	Damped and bounded operational-state factor	–	42	
Corrections, energy, and carbon	$P_{tot,t}$	Final total electrical power	kW	43	
Corrections, energy, and carbon	E_t	Electrical energy consumed during one simulation period	kWh	43–44, 49, 59	
Corrections, energy, and carbon	$MCO_{2,t}$	Operational carbon emissions associated with electricity use	kg CO ₂ e	44, 49, 51	
Corrections, energy, and carbon	EF_{grid}	Electricity-grid emission factor	kg CO ₂ e kWh ⁻¹	44, 49, 51	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
Comfort indicator	$T_{z,ind,t}$	Empirical zone-temperature indicator used by the current objective	°C	45, 47	
Comfort indicator	$k_{RH,c}$	Humidity contribution coefficient in the empirical comfort indicator	°C %-point ⁻¹	45	
Comfort indicator	$P_{cap,t}$	Empirical unmet-capacity temperature penalty	°C	45–46	
Comfort indicator	$Q_{unmet,t}$	Thermal load exceeding available HVAC capacity	kW	46	
Comfort indicator	T_{set}	Nominal comfort reference temperature	°C	47, 52	
Comfort indicator	$DT_{,t}$	Absolute empirical comfort deviation	K	47, 49	
S3 optimization	x_t	S3 control vector $[T_{sp,t}, \alpha_t]^T$	mixed	48	
S3 optimization	$T_{sp,min}, T_{sp,max}$	Lower and upper setpoint bounds	°C	48, 53	
S3 optimization	$\alpha_{min}, \alpha_{max}$	Lower and upper airflow-fraction bounds	–	48, 53	
S3 optimization	$\bar{e}_t$	Recommended notation for normalized energy objective; replaces e_t only typographically	–	49–51	
S3 optimization	$\bar{\delta}_t$	Recommended notation for normalized degradation objective; equal to δ_t in the current code	–	49–51	
S3 optimization	$\bar{D}_t$	Recommended notation for normalized comfort-deviation objective	–	49–51	
S3 optimization	$\bar{M}_t$	Recommended notation for normalized carbon objective	–	49–51	
S3 optimization	w_E, w_D, w_C, w_G	Weights assigned to energy, degradation, comfort, and carbon terms	–	50–51	
S3 optimization	J_t	Weighted S3 objective-function value	–	50–51	
S3 optimization	μ_k	Mean/search-centre vector of the Gaussian candidate distribution	mixed	52–55	
S3 optimization	σ_k	Standard-deviation vector of the Gaussian candidate distribution	mixed	52–55	
S3 optimization	$\varepsilon_{k,i}$	Standard-normal random candidate vector	–	53	

Category	Symbol	Definition	SI unit	Equation(s)	Implementation note
S3 optimization	$x_{k,i}$	Candidate control vector i at iteration k	mixed	53, 55	
S3 optimization	N_p	Candidate population size	candidate	54–55	
S3 optimization	N_e	Number of elite candidates	candidate	54–55	
S3 optimization	N_{iter}	Fixed number of stochastic-search iterations	iteration	55	
Maintenance and cost	$u_{HX,t}$	Binary heat-exchanger cleaning decision	0 or 1	56, 58, 60	
Maintenance and cost	$u_{F,t}$	Binary filter-replacement decision	0 or 1	57–58, 60	
Maintenance and cost	$R_{f,warn}$	Heat-exchanger fouling warning threshold	$m^2 K W^{-1}$	56	
Maintenance and cost	ΔP_{warn}	Filter pressure-drop warning threshold	Pa	57	
Maintenance and cost	δ_{trig}	Composite degradation maintenance trigger	–	56–57	
Maintenance and cost	CE_t	Period electricity cost	USD	59, 61	
Maintenance and cost	p_E	Electricity tariff	USD kWh ⁻¹	59	
Maintenance and cost	CM_t	Direct maintenance cost during the period	USD	60–61	
Maintenance and cost	CF	Unit cost of filter replacement	USD action ⁻¹	60	
Maintenance and cost	CHX	Unit cost of heat-exchanger cleaning	USD action ⁻¹	60	
Maintenance and cost	$C_{tot,t}$	Total period energy plus maintenance cost	USD	61	

Important interpretation: The nomenclature table defines symbols, but it does not convert empirical coefficients into literature constants. Coefficients and correction arrays that originate from the code must be reported as calibrated or assumed model parameters.

2. Equation-to-reference map

Equation range	Supporting reference(s)	Honest attribution statement
1–18	ASHRAE (2021); U.S. Department of Energy (2024)	Physical structure is literature-based; reduced-order coefficients and switches are study-specific and require calibration/validation.
19–25	Bacher and Madsen (2011); U.S. Department of Energy (2024)	RC/state-space concept is supported; the specific two-state discretization, blending fraction, and parameter values are the present model.

Equation range	Supporting reference(s)	Honest attribution statement
26–29	No external equation source	Empirical monthly availability and minimum-load calibration developed for this study. Report and release all arrays.
30	Kern and Seaton (1959)	Classical asymptotic fouling form.
31–33	ASHRAE (2017); Teng et al. (2022)	Dust-loading/pressure-drop relationship is supported; linear dust growth, kclog, and the composite index are study-specific.
34–35	U.S. Department of Energy (2024); ASHRAE (2021)	Temperature/COP performance-curve concept is supported. Coefficients 0.45, 0.30, 0.018, and 0.010 are empirical model assumptions, not standard coefficients.
36–40	ASHRAE (2021); U.S. Department of Energy (2024)	Power-balance structure is conventional; monthly floors and degradation multipliers are study-specific.
41–42	No external equation source	Explicit empirical calibration layers. They must not be described as universal physics.
43	ASHRAE (2021)	Power-to-energy integration is a dimensional identity.
44	World Resources Institute and World Business Council for Sustainable Development (2015)	Electricity use multiplied by a grid emission factor.
45–47	ASHRAE (2023), terminology only	Current equation is an empirical indicator, not PMV, PPD, operative temperature, or Standard 55 compliance.
48–51	Marler and Arora (2010)	Weighted-sum formulation; current carbon normalization duplicates the energy term under constant EFgrid.
52–55	Rubinstein (1999), conceptual only	The code is cross-entropy-like but not a formal implementation of the published cross-entropy algorithm.
56–58	Jardine et al. (2006), classification only	Threshold logic and complete-restoration assumptions are study-specific.
59–61	Standard engineering cost accounting	Direct algebraic accounting; tariffs and action costs are user/model inputs.

3. APA 7th edition references

American Society of Heating, Refrigerating and Air-Conditioning Engineers. (2021). 2021 ASHRAE handbook—Fundamentals. ASHRAE.

Use in this manuscript: General basis for building loads, heat transfer, infiltration, internal gains, fan/pump concepts, and HVAC design quantities (Eqs. 1–18 and 36–40).

American Society of Heating, Refrigerating and Air-Conditioning Engineers. (2017). ANSI/ASHRAE Standard 52.2-2017: Method of testing general ventilation air-cleaning devices for removal efficiency by particle size. ASHRAE.

Use in this manuscript: Supports the physical relationship among filter dust loading, pressure-drop increase, and final filter resistance. The linear coefficients in Eqs. 31–33 remain study-specific simplifications.

American Society of Heating, Refrigerating and Air-Conditioning Engineers. (2023). ANSI/ASHRAE Standard 55-2023: Thermal environmental conditions for human occupancy. ASHRAE.

Use in this manuscript: Used only to delimit terminology: Eqs. 45–47 are an empirical comfort indicator and are not a Standard 55 PMV/PPD compliance calculation.

Bacher, P., & Madsen, H. (2011). Identifying suitable models for the heat dynamics of buildings. *Energy and Buildings*, 43(7), 1511–1522.
<https://doi.org/10.1016/j.enbuild.2011.02.005>

Use in this manuscript: Primary basis for lumped RC/grey-box thermal-state modelling (Eqs. 19–25).

Jardine, A. K. S., Lin, D., & Banjevic, D. (2006). A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing*, 20(7), 1483–1510. <https://doi.org/10.1016/j.ymssp.2005.09.012>

Use in this manuscript: Supports the classification of S3 as condition-based maintenance based on current condition indicators (Eqs. 56–58), rather than forecast-based predictive maintenance.

Kern, D. Q., & Seaton, R. E. (1959). A theoretical analysis of thermal surface fouling. *British Chemical Engineering*, 4(5), 258–262.

Use in this manuscript: Classical asymptotic fouling-growth basis for Eq. 30.

Marler, R. T., & Arora, J. S. (2010). The weighted sum method for multi-objective optimization: New insights. *Structural and Multidisciplinary Optimization*, 41(6), 853–862.
<https://doi.org/10.1007/s00158-009-0460-7>

Use in this manuscript: Weighted-sum objective formulation and interpretation of objective weights (Eqs. 49–51).

Rubinstein, R. Y. (1999). The cross-entropy method for combinatorial and continuous optimization. *Methodology and Computing in Applied Probability*, 1(2), 127–190.
<https://doi.org/10.1023/A:1010091220143>

Use in this manuscript: Conceptual basis for iterative sampling, elite selection, and distribution updating. The implemented routine is only cross-entropy-like and should be named an elite Gaussian stochastic search (Eqs. 52–55).

Teng, G., Shi, G., & Zhu, J. (2022). Influence of pleated geometry on the pressure drop of filters during dust loading process: Experimental and modelling study. *Scientific Reports*, 12, 20331. <https://doi.org/10.1038/s41598-022-24838-7>

Use in this manuscript: Experimental/model evidence that pressure drop evolves with deposited dust mass. The simple linear form in Eq. 32 is a reduced-order assumption requiring calibration.

U.S. Department of Energy. (2024). EnergyPlus engineering reference (Version 24.1). EnergyPlus.

Use in this manuscript: Reference framework for zone heat balance, thermostat load calculations, fan power, and performance-curve HVAC modelling (Eqs. 12–25 and 34–40).

World Resources Institute, & World Business Council for Sustainable Development. (2015). GHG Protocol Scope 2 guidance: An amendment to the GHG Protocol corporate standard. World Resources Institute.

Use in this manuscript: Basis for calculating electricity-related operational emissions as activity data multiplied by an emission factor (Eq. 44).

Citation warning. Do not cite the empirical monthly factors, operational-state factors, comfort coefficients, or exact COP-degradation coefficients as though they came from the references above. Those quantities are model-specific and require calibration, sensitivity analysis, and validation.